Part 1: Data Exploration and Visualization (20 marks)

Q1 (5 marks) Load the dataset lifesat.csv and display the first 5 rows.

In [35]:
```
import pandas as pd
import matplotlib.pyplot as plt;
import seaborn as sns;
import warnings
warnings.filterwarnings("ignore")

path = r"C:\pp\ml--2026\5w\lifesat.csv"
df = pd.read_csv(path, sep = ",")

df.head(5)
```

Out[35]:

	Country	GDP per capita (USD)	Life satisfaction
0	Russia	26456.387938	5.8
1	Greece	27287.083401	5.4
2	Turkey	28384.987785	5.5
3	Latvia	29932.493910	5.9
4	Hungary	31007.768407	5.6

Q2 (5 marks) Print basic info and summary statistics .

In [36]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27 entries, 0 to 26
Data columns (total 3 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Country               27 non-null     object 
 1   GDP per capita (USD)  27 non-null     float64
 2   Life satisfaction     27 non-null     float64
dtypes: float64(2), object(1)
memory usage: 780.0+ bytes
```

In [37]:
```
df.describe()
```

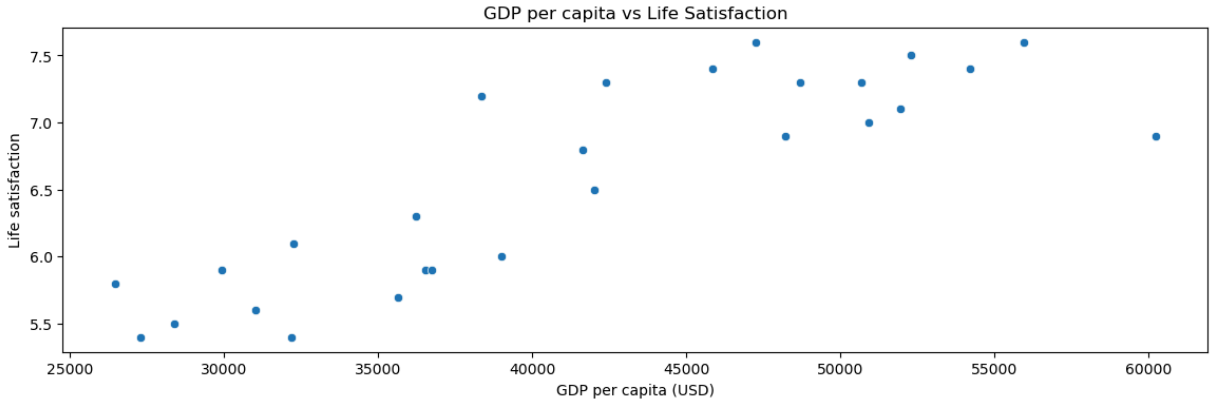
Out[37]:

	GDP per capita (USD)	Life satisfaction
count	27.000000	27.000000
mean	41564.521771	6.566667
std	9631.452319	0.765607
min	26456.387938	5.400000
25%	33938.289305	5.900000
50%	41627.129269	6.800000
75%	49690.580269	7.300000
max	60235.728492	7.600000

Q3 (10 marks) Display a scatter plot for GDP per capita vs Life Satisfaction . Add labels, title, and discuss the observed relationship

In [38]:
```
plt.figure(figsize = (14,4))

plt.title("GDP per capita vs Life Satisfaction")
sns.scatterplot(x = "GDP per capita (USD)" , y = "Life satisfaction", data = df)
plt.show()
```



In [39]:
```
# The scatter plot shows a clear positive relationship between GDP per capita and l
# As GDP per capita increases, life satisfaction also tends to increase.
# The relationship appears approximately linear, although the increase in life sati
# There are no strong outliers, and the overall pattern suggests that higher econom
```

Part 2: Linear Regression Model (30 marks)

Q4 (5 marks) Extract input (X) and target (y). Print their shapes.

In [40]:
```
X = df[["GDP per capita (USD)"]]
y = df["Life satisfaction"]

print("X shape:", X.shape)
print("y shape:", y.shape)
```

```
X shape: (27, 1)
y shape: (27,)
```

Q5 (10 marks) Train a Linear Regression model & Display coefficient and intercept.

In [41]:
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X,y)

print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
```

```
Coefficient: 6.778899694341219e-05
Intercept: 3.74904942737691
```

Q6 (10 marks) Plot the predicted regression line from the model along with a scatter plot of the data.

In [42]:
```
plt.figure(figsize=(8,5))

plt.title("Linear Regression Fit")
sns.regplot(x="GDP per capita (USD)",
    y="Life satisfaction",
    data=df,
    ci=None
           )
plt.show()

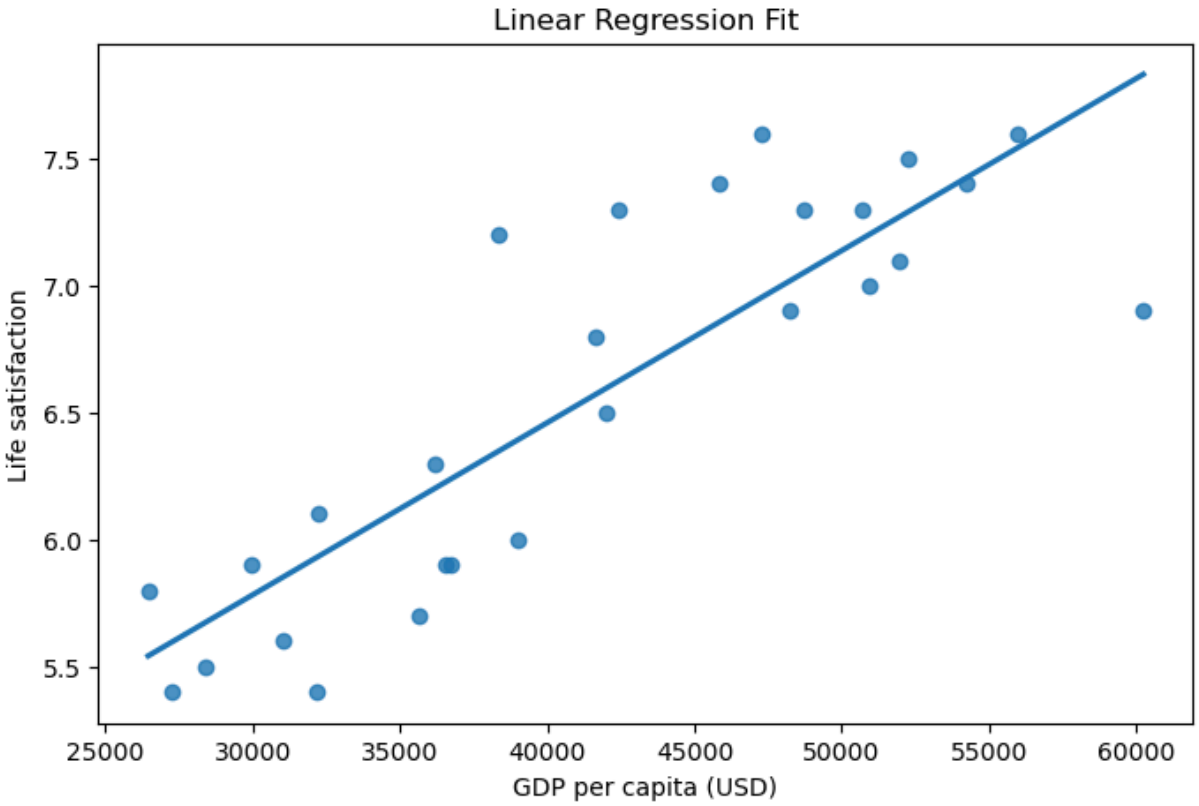
# or

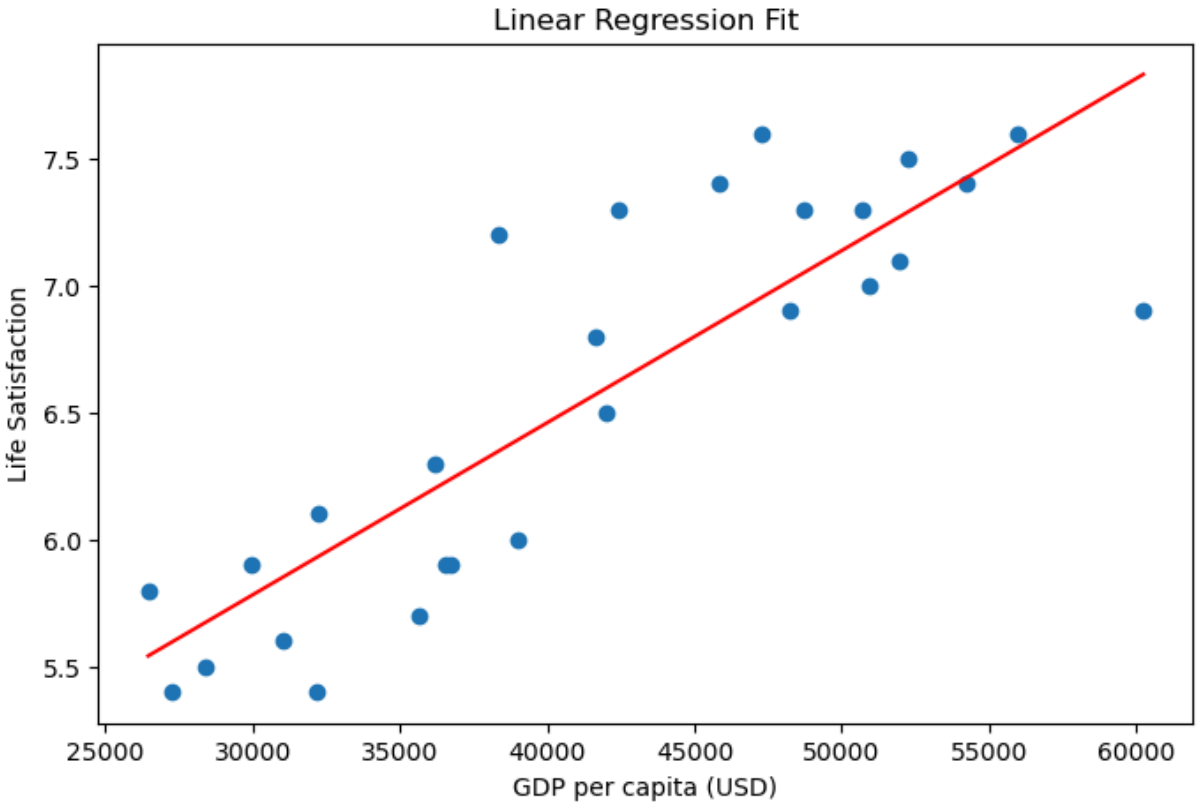
plt.figure(figsize=(8,5))
plt.scatter(X, y)
y_pred = model.predict(X)

#Line
plt.plot(X, y_pred, color='red')

plt.xlabel("GDP per capita (USD)")
plt.ylabel("Life Satisfaction")
plt.title("Linear Regression Fit")

plt.show()
```





Q7 Predict for GDP = 37655.2

In [43]:
```
gdp_value = [[37655.2]]
prediction = model.predict(gdp_value)
```

```
print("Predicted Life Satisfaction:", prediction[0])
```

```
Predicted Life Satisfaction: 6.3016576650804845
```

In [44]:
```
# The predicted life satisfaction for a GDP per capita of 37,655.2 USD is approxima
# This value is consistent with the positive relationship observed in the data,
# indicating that countries with mid-range GDP levels tend to have moderate life sa
```

Part 3: K-Nearest Neighbors Regression (25 marks)

Q8 (5 marks) Train a KNeighborsRegressor (n_neighbors=3).

In [45]:
```
from sklearn.neighbors import KNeighborsRegressor

knn = KNeighborsRegressor(n_neighbors=3)
knn.fit(X, y)
```

Out[45]:

KNeighborsRegressor

▶ Parameters

Q9 Predict Life Satisfaction for GDP = 37,655.2 USD and compare with Linear Regression.

In [47]:
```
gdp_value = [[37655.2]]
knn_pred = knn.predict(gdp_value)

print("KNN Prediction:", knn_pred[0])


#The KNN prediction is slightly different from the Linear Regression result.
# While Linear Regression assumes a global linear relationship, KNN uses local aver
# Therefore, KNN may capture local variations in the data, leading to slightly diff
```

```
KNN Prediction: 6.333333333333333
```

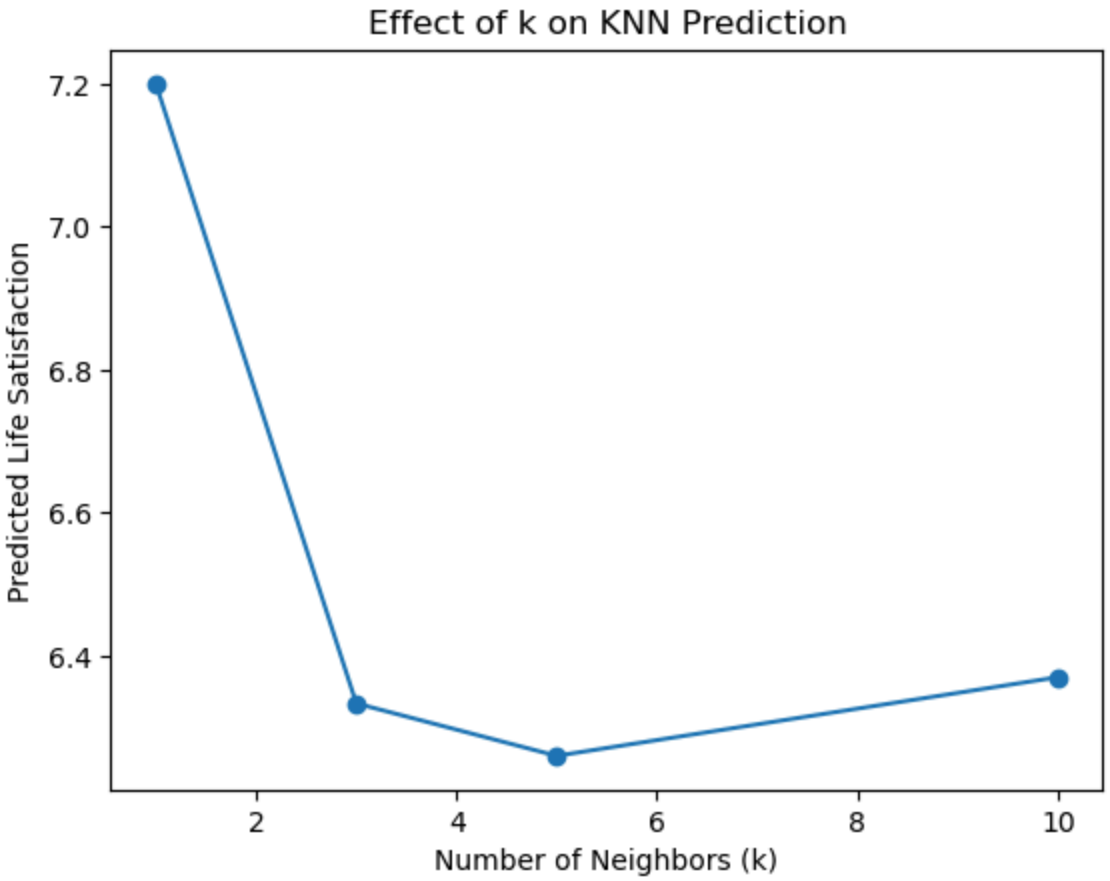
Q10 Use n_neighbors 1, 3, 5, and 10 and print the predicted values of life satisfaction in Q10. Plot the results using a line plot.

In [48]:
```
k_values = [1, 3, 5, 10]
predictions = []

for k in k_values:
    model_knn = KNeighborsRegressor(n_neighbors=k)
    model_knn.fit(X, y)
    pred = model_knn.predict([[37655.2]])
    predictions.append(pred[0])
    print(f"k={k} → Prediction: {pred[0]}")
```

```
k=1 → Prediction: 7.2
k=3 → Prediction: 6.333333333333333
k=5 → Prediction: 6.26
k=10 → Prediction: 6.369999999999999
```

In [49]:
```
plt.plot(k_values, predictions, marker='o')
plt.xlabel("Number of Neighbors (k)")
plt.ylabel("Predicted Life Satisfaction")
plt.title("Effect of k on KNN Prediction")
plt.show()
```



In [50]:
```
# As k increases, the prediction becomes smoother and less sensitive to individual
# Small k values may lead to overfitting, while larger k values produce more stable
```

In [ ]:
```

```